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Jongho Kim

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Cornell Johnson Graduate School of Management | https://jonghkim.github.io

Education

Cornell University, SC Johnson Graduate School of Management Aug 2021–Present
Ph.D. in Quantitative Marketing
Minor Field: Applied Statistics
Committee: Vrinda Kadiyali (Chair), Emaad Manzoor, Omid Rafieian, Dan Kowal (Statistics)

Korea Advanced Institute of Science and Technology Feb 2015–Feb 2017
M.S. in Management Engineering, Information Systems

Ulsan National Institute of Science and Technology Mar 2010–Aug 2014
B.S. in Computer Science and Engineering, Minor in Finance and Accounting

Journal Publications

Park, J. and **Kim, J.**, 2022. "A Data-Driven Exploration of the Race between Human Labor and Machines in the 21st Century," *Communications of the ACM* [Manuscript] [SSRN] [Website] [Video] [Code]

Choi, S., Kim, W. and **Kim, J.**, 2016. "Compression of Hamiltonian Matrix: Application to Spin-1/2 Heisenberg Square," *AIP Advances* (**Editor's Pick**) [Manuscript]

Working Papers

Kim, J. and Kadiyali, V., "Estimating Treatment Effects with Large Language Models: Evidence from San Diego's Algorithmic Pricing Ban" **(Job Market Paper)**

Kim, J., Kowal, D., and Ruppert, D., "Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors"

Kim, J., Anand, P., and Kadiyali, V., "The Effects of Rent Control on Renter Experience and Consumption"

Selected Conference and Seminar Presentations

Kim, J., Kowal, D., and Ruppert, D., 2026, "Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors," *IO / Empirical Modeling Brown Bag*, Cornell University, Ithaca, NY, USA

Kim, J., Anand, P., and Kadiyali, V., 2025, "The Effects of Rent Control on Consumer Reviews on Rental Housing," *ISMS Marketing Science Conference*, Washington, D.C., USA

Park, J. and **Kim, J.**, 2018. "Leveraging Machine Learning to Reduce Racial Bias on Online Platforms: A Neural Machine Translation Approach," *INFORMS Conference on Information Systems and Technology (CIST)*, Phoenix, Arizona, USA

Kim, J., Cho, D. and Lee, B., 2016. "The Mind Behind Crowdfunding: An Empirical Study of Speech Emotion in Fundraising Success," *International Conference on Information Systems (ICIS)*, Dublin, Ireland (**Most Innovative Research-in-Progress Paper Runner-up**) [Manuscript] [Poster]

Park, J., **Kim, J.** and Lee, B., 2016. "Which Tasks Will Technology Take? A New Systematic Methodology to Measure Task Automation," *International Conference on Information Systems (ICIS)*, Dublin, Ireland [Manuscript] [Poster]

Professional Experience

December & Company Asset Management, Seoul, South Korea Apr 2019–Jul 2021
Senior Researcher / Quantitative Portfolio Manager, Portfolio Research Team

- Develop Generative Models for Textual Factors to Predict and Explain Asset Returns based on SEC-10X (10-K, 10-Q, 10-KSB and 10-QSB)
- Implement a Sector Rotation Strategy through Cross-Sectional Variation over Data-driven Economic Cycles
- Construct Alternative Investment Strategies, especially for Commodity, REITs, Infrastructure and High Dividend-Paying ETFs
- Work as a Substitute for Mandatory Military Service

NICE Pricing and Information, Seoul, South Korea Aug 2016–Apr 2019
Quantitative Researcher, Financial Engineering Center

- Develop Non-Homogeneous Time Markov Chain Models (NHCTMC) for Forward Looking Probability of Default under IFRS9 based on Sparse Transition Matrix of Credit Ratings
- Estimate Maximum Likelihood Proxy Portfolio of Fixed-Income Portfolio via Return based Style Analysis (RBSA)
- Implement Implied Volatility Surface of Index Futures Options via Heston Stochastic Volatility Model
- Work as a Substitute for Mandatory Military Service

Digital Economics and Enterprise Research Lab, KAIST, Seoul, Korea May 2015–Feb 2017

- Empirical Research in Information Systems using Machine Learning and Econometrics (Advisor: Dr. Byungtae Lee)
- Accepted Three Conference Papers in ICIS 2016 and Awarded for the Most Innovative Research-in-Progress Paper Award (Runner-Up) in ICIS 2016
- Government & Industry Projects
  Strategies for Online to Offline (O2O) Transport Services, The Korea Transport Institute (Advisor: Dr. Lee, B.), Sep 2016–Nov 2016
  The Effect of Model-Driven Development (MDD) on Information Systems Maintenance, LG CNS (Advisor: Dr. Lee, B.), Sep 2015–Dec 2015

Computational Mathematical Science Lab, UNIST, Ulsan, Korea Sep 2011–Feb 2015

- Research and Study for Complex Network Theory and Numerical Methods for Differential Equations
- Published a Paper in AIP Advances (Compressed Representation of the Hamiltonian Matrix) and Selected as an Undergraduate Research Opportunity Program (Optimal Control Strategies of Lotka–Volterra Equation)

NewsJAM, Ulsan, South Korea Jun 2013–Feb 2014

Chief Technology Officer (CTO), Co-Founder

- Developed NLP engines for news analysis including text summarization, key sentence extraction, topic modeling, and text clustering [PyCon Korea 2016]
- Open SW Incubator ($60,000 as award), National IT Industry Promotion Agency, 2013

Honors & Awards

AMA-Sheth Doctoral Consortium Fellow 2026

ISMS Doctoral Consortium Fellow 2025

FinTech@Cornell Student Fellowship 2022

Cornell SC Johnson College Doctoral Fellowship 2021–current

Best Student Paper Award, Post-ICIS 2017 KrAIS Research Workshop 2017

Best Graduate Student Paper Award, KAIST 2017

Most Innovative Research-in-Progress Paper Runner-Up, International Conference on Information Systems (ICIS), Dublin, Ireland 2016

Editor's Picks from the American Institute of Physics, AIP Advances 2016

Scholarship for Academic Excellence among 2015 Entering Class, KAIST 2015

Excellence Prize (Top 1%, 3rd Place over 322 teams) at Hackathon for Big Data, Ministry of Science, ICT and Future Planning 2014

Dean's List for Four Semesters, UNIST 2010, 2012–2014

Undergraduate Research Opportunity Program, Korea Foundation for Advancement of Science & Creativity 2013

National Natural Science and Engineering Scholarship, Korea Student Aid Foundation 2010–2014

Patents

Kim, J., et al., "Automated Trading System and Method," Korean Patent 10-2433324, Assignee: December & Company Inc., 2022. (Application No. 10-2019-0117628, filed Sep 24, 2019)

Teaching

Instructor

Cornell University Fall 2025

BANA 5440 Introduction to Data Programming, MSBA

StudyPie Oct 2018–Apr 2019

Deep Learning Models for Financial Time Series Prediction [Link]

Guest Lecturer

KAIST FMBA – Introduction to FinTech (Prof. Lee, B.) Mar 2021, Mar 2023, Apr 2024
Investment Strategies in FinTech Industry [Link]

Korea Summer Session on Causal Inference (Prof. Park, J.) Jul 2021
Applications of Machine Learning in Marketing [Link]

KAIST MBA – Business Analytics (Prof. Oh, W.) Apr 2019
Intelligent Video Analytics with Deep Learning [Link]

KAIST IT Management Summer Research Workshop (Prof. Park, J.) Aug 2017
Machine Learning APIs and Web Scraping [Link]

Teaching Assistant

Cornell University

NCCT 5030/5031 Marketing Management Spring, Fall 2023
AEM 4435 / NBA 6390 Data-Driven Marketing Fall 2022
NCC 5030 Marketing Management Summer 2022

Invite-Only Workshops

Econometric Society, Africa Training Workshop in Econometrics, 2024

Services

Ad-hoc Reviewer

International Conference on Information Systems (ICIS), 2021
International Conference on Information Systems (ICIS), 2019
International Conference on Electronic Commerce (ICEC), 2017

Other Information

Work Authorization: U.S. Permanent Residence (I-485 Pending); EAD Expected July 2026; No Sponsorship Required
Language: Korean (Native), English (Fluent)
Programming: Python, C/C++, Java, R, STATA, Matlab, SQL, JavaScript, HTML, CSS

References

Vrinda Kadiyali (Chair)
Nicholas H. Noyes Professor of Management
Professor of Marketing and Economics
SC Johnson Graduate School of Management
Cornell University
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Omid Rafieian
Assistant Professor of Marketing
Demir Sabanci Faculty Fellow
Cornell Tech and SC Johnson College of Business
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Emaad Manzoor
Assistant Professor of Marketing
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Dan Kowal
Associate Professor of Statistics and Data Science
Bowers College of Computing & Information Science
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Selected Research Abstracts

"Estimating Treatment Effects with Large Language Models: Evidence from San Diego's Algorithmic Pricing Ban," Kim, J. and Kadiyali, V. (Job Market Paper)

How can we use LLM outputs for causal inference when they are imperfect rather than ground truth? LLMs read constructs from text at scale, but they hallucinate mentions that are absent, omit ones that are present, and miscalibrate the values they report. We develop a semiparametric Bayesian framework that estimates treatment effects on latent satisfaction from a discrete rating by combining a small labeled sample with large-scale unlabeled LLM outputs and jointly estimating the latent outcome, the measurement model, and the treatment effect. We apply it to San Diego's 2025 algorithmic-pricing ban, which bars setting rents from competitors' nonpublic data, across 58,293 Google Maps reviews of 1,686 apartments. Nine months after the ban, renters rate their apartments 0.15 points lower on a 5-point scale, counter to its pro-renter intent. In the text, satisfaction with rent price falls 0.26 while non-price services rise 0.37. The ban appears to raise pricing inefficiency and shift landlords from price to service competition, leaving renters worse off. As the DOJ–RealPage settlement takes the same regime nationwide, our findings inform how policymakers weigh its consequences.

"The Effects of Rent Control on Renter Experience and Consumption: Evidence from St. Paul's 2022 Ordinance," Kim, J., Anand, P., and Kadiyali, V.

How does housing affordability policy reshape consumer experience and grocery spending? We examine the impact of St. Paul's 2022 rent control policy using 110,258 renter reviews from 2,285 apartments and grocery sales data from 123 stores in 18 months post-policy. We compare St. Paul (policy area) and surrounding spillover areas to comparable U.S. cities as controls. Renters rate their apartment experiences higher by 0.474 points (on a 5-point scale) in the policy area and by 0.157 points in the spillover areas. In the 18 months after policy implementation, no dimension of the multi-attribute service experience worsens, suggesting net satisfaction gains for renters in our time window. Grocery expenditure declines by 8% in the policy area and 3% in the spillover areas over the two years following implementation through different mechanisms: the exit of higher-income renters from the policy area and likely increased rents in the spillover areas. This reveals a policy tradeoff: improved renter satisfaction dampened grocery demand in affected neighborhoods. Our findings have implications for apartment managers to optimize service quality and consumer retention under price regulation; grocery retailers to recalibrate assortment and pricing for shifting neighborhood demographics; and policymakers to weigh the full consequences of rent control across renters, businesses, and spillover areas.

"Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors," Kim, J., Kowal, D., and Ruppert, D.

Large language models (LLMs) enable researchers to measure constructs of interest from text at scale. We study treatment effect estimation using LLM-measured outputs as causal outcomes. Standard plug-in estimators can substantially underestimate uncertainty, even when point estimates are unbiased. An LLM's errors are not independent. It misreads semantically similar units in similar ways, inducing correlated errors. This correlation is information, not nuisance, but its structure is unknown a priori. We develop a Bayesian framework that learns this dependence from embedding geometry with a Gaussian process, estimating it jointly with the calibration model and the treatment effect using a small labeled sample together with a large set of unlabeled LLM outputs. Estimating jointly, rather than in stages, allows labeled and unlabeled data to inform each other, while the learned covariance structure induces precision-weighted posterior inference, improving statistical efficiency over methods that assume independent errors. Together, these restore valid inference and improve precision over methods that assume independent errors. In a randomized experiment, we document that this dependence is real, survives fine-tuning, and escapes conventional clustering, and across simulations our framework maintains valid coverage while improving efficiency. The results show that dependence among an LLM's errors can be exploited rather than ignored.